Observing that the sum of ℓ independent Guassian random variables of variance c_k is a Gaussian random variable with variance $\sum_{k=1}^{\ell} c_k$, we see that this is in the form of Langevin dynamics

$$\mathrm{d}p_i = F_i \mathrm{d}t - \hat{\gamma} p_i \mathrm{d}t + \sqrt{2\mathrm{k_B T}\hat{\gamma}}\mathrm{dW}_{i,k},$$

with

$$\hat{\gamma} = \sum_{k=1}^{N_k} c_k.$$

(For the formal treatment of this limit refer to [293].)

For numerical integration, the system (8.68)–(8.70) can be handled via splitting. A natural family of methods based on Verlet is the following [19]:

Generalized Langevin Dynamics Method

$$\begin{aligned}
\boldsymbol{p}_{n+1/2} &:= \boldsymbol{p}_n + \frac{h}{2}\boldsymbol{F}(\boldsymbol{q}_n) + \frac{h}{2}\sum_{k=1}^{k_{\max}} \boldsymbol{\sigma}_{n,k}, \\
\boldsymbol{q}_{n+1} &:= \boldsymbol{q}_n + h\boldsymbol{M}^{-1}\boldsymbol{p}_{n+1/2}, \\
\boldsymbol{\sigma}_{n+1,k} &:= \theta_k \boldsymbol{\sigma}_{n,k} - (1-\theta_k)\, c_k \boldsymbol{p}_{n+1/2} + \alpha_k \sqrt{2\mathrm{k_B T} c_k}\, \boldsymbol{Z}_k^n, \\
\boldsymbol{p}_{n+1} &:= \boldsymbol{p}_{n+1/2} + \frac{h}{2}\boldsymbol{F}(\boldsymbol{q}_{n+1}) + \frac{h}{2}\sum_{k=1}^{k_{\max}} \boldsymbol{\sigma}_{n+1,k},
\end{aligned}$$

where $\boldsymbol{\sigma}_{n,k}$ is a $\mathrm{N_c}$-dimensional vector, and the components of the $\mathrm{N_c}$-dimensional vectors $\mathbf{Z}_{n,k}$ are independent standard normal random variables.

The real-valued coefficients θ_k and α_k which must satisfy, for consistency, the equations

$$\theta_k = 1 - \frac{h}{\tau_k} + \mathcal{O}(h^2), \qquad \alpha_k = \frac{\sqrt{h}}{\tau_k} + \mathcal{O}(h).$$

An argument is given in [19] that α_k and θ_k should be taken as

$$\theta_k = \exp(-h/\tau_k), \qquad \alpha_k = \sqrt{\frac{(1-\theta_k)^2}{h}},$$

since, for this choice, the Langevin limit is preserved. Alternative methods are possible. For many comparisons and numerical experiments, refer to [19].